

100 Linux Troubleshooting Scenarios

Real-World Diagnosis & Fix Guide

DevOps Shack

devopsshack.com | @devopsshack

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BOOT & SYSTEM STARTUP — Real-World Troubleshooting

#1 System Fails to Boot — GRUB Rescue Prompt

SYMPTOM	System drops to grub rescue> prompt instead of booting OS.
ROOT CAUSE	GRUB config corrupted, boot partition deleted, or wrong root partition UUID in grub.cfg.
FIX	Set root and prefix manually in GRUB rescue, then boot and run grub-install + update-grub.
<pre>grub rescue> set root=(hd0,1) grub rescue> set prefix=(hd0,1)/boot/grub grub rescue> insmod normal && normal</pre>	

#2 Kernel Panic on Boot

SYMPTOM	System shows Kernel panic - not syncing and halts.
ROOT CAUSE	Corrupt initramfs, missing kernel module, or hardware failure (bad RAM/disk).
FIX	Boot from rescue/live USB. Rebuild initramfs: mkinitcpio -P or update-initramfs -u.
<pre>mkinitcpio -P # Arch update-initramfs -u -k all # Debian/Ubuntu</pre>	

#3 System Stuck at fsck on Boot

SYMPTOM	Boot hangs at: /dev/sda1: UNEXPECTED INCONSISTENCY; RUN fsck MANUALLY.
ROOT CAUSE	Filesystem not cleanly unmounted; forced fsck required.
FIX	Run fsck from single-user or rescue mode. Answer yes to prompts or use -y flag.
<pre>fsck -y /dev/sda1</pre>	

#4 systemd Service Failing to Start on Boot

SYMPTOM	Service shows failed in systemctl status; system enters degraded state.
ROOT CAUSE	Missing dependency, wrong ExecStart path, permission error, or misconfigured unit file.
FIX	journalctl -xe to see full failure reason. Fix unit file, verify binary path, check deps.
<pre>systemctl status <service> journalctl -xe -u <service></pre>	

#5 Boot Dropping to Emergency Mode

SYMPTOM	System shows You are in emergency mode after failed mount.
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ROOT CAUSE	Bad entry in /etc/fstab — wrong UUID or unmountable device.
FIX	Comment out bad fstab entry. Run blkid to find correct UUIDs. Fix /etc/fstab.
	<code>blkid && nano /etc/fstab</code>

#6 Slow Boot — Long systemd Startup Time

SYMPTOM	Boot takes 2+ minutes; systemd-analyze shows one service dominating.
ROOT CAUSE	Single slow service (often network-online.target, cloud-init, or NFS mount) blocking boot.
FIX	systemd-analyze blame to find culprit. Disable or fix slow service.
	<code>systemd-analyze blame head -20 systemd-analyze critical-chain</code>

#7 initramfs Prompt on Boot

SYMPTOM	Boot drops to initramfs (initramfs) prompt: ALERT! /dev/disk/by-uuid/... does not exist.
ROOT CAUSE	Root device UUID mismatch in GRUB config vs actual disk UUID (after disk replacement/clone).
FIX	Update /boot/grub/grub.cfg or /etc/default/grub with correct UUID from blkid.
	<code>blkid /dev/sda1 update-grub</code>

#8 Wrong Default Boot Entry in GRUB

SYMPTOM	System always boots old kernel; new kernel never selected.
ROOT CAUSE	GRUB_DEFAULT set to fixed index instead of 'saved'; update-grub not run after kernel install.
FIX	Set GRUB_DEFAULT=saved, GRUB_SAVEDEFAULT=true in /etc/default/grub, then update-grub.
	<code>grep GRUB_DEFAULT /etc/default/grub update-grub</code>

#9 System Time Wrong After Boot

SYMPTOM	System clock shows wrong time, causing certificate errors and log confusion.
ROOT CAUSE	RTC clock set to local time but systemd expects UTC, or NTP not syncing.
FIX	Set RTC to UTC: timedatectl set-local-rtc 0. Enable NTP: timedatectl set-ntp true.
	<code>timedatectl set-ntp true timedatectl status</code>

#10 Plymouth Splash Screen Causing Boot Hang

SYMPTOM	Boot stalls at splash screen indefinitely; removing quiet splash fixes it.
ROOT CAUSE	Plymouth framebuffer driver conflict with GPU driver or display hardware.
FIX	Remove quiet splash from GRUB_CMDLINE_LINUX_DEFAULT in /etc/default/grub.
	<pre>sed -i 's/quiet splash//' /etc/default/grub && update-grub</pre>

#11 Multi-User Target Not Reached

SYMPTOM	System stops at basic.target or rescue.target; not all services start.
ROOT CAUSE	A required service fails and prevents subsequent targets from being activated.
FIX	journalctl -b to read boot log. Identify failed unit and fix or mask it.
	<pre>journalctl -b -p err systemctl list-units --state=failed</pre>

#12 Dual Boot Windows Time Conflict

SYMPTOM	After switching between Linux and Windows, system clock is wrong in one OS.
ROOT CAUSE	Windows uses local time for RTC; Linux uses UTC; they overwrite each other.
FIX	Make Linux use local RTC to match Windows: timedatectl set-local-rtc 1.
	<pre>timedatectl set-local-rtc 1 --adjust-system-clock</pre>

DISK & FILESYSTEM — Real-World Troubleshooting

#13 Disk Full — No Space Left on Device

SYMPTOM	Write operations fail: No space left on device; df -h shows 100% usage.
ROOT CAUSE	Logs, core dumps, large files, or deleted-but-open files consuming all space.
FIX	du -sh /* to find large dirs. lsof +L1 for deleted-but-open files. Clean logs/tmp.
	<pre>df -h && du -sh /var/log/* sort -rh head -20 lsof +L1</pre>

#14 Inode Exhaustion — Disk Full But df Shows Free Space

SYMPTOM	Cannot create new files; df -h shows space free but df -i shows 100% inode usage.
ROOT CAUSE	Millions of small files (cache, mail queue, temp files) exhausting inode table.
FIX	Find inode hogs: find / -xdev -printf '%h ' sort uniq -c sort -rn head. Delete or archive.
	<pre>df -i && find /tmp -xdev -printf '%h ' sort uniq -c sort -rn head -10</pre>

#15 Filesystem Mounted Read-Only Unexpectedly

SYMPTOM	Cannot write to filesystem; mount shows ro; kernel log shows I/O errors.
ROOT CAUSE	Disk I/O errors triggered kernel to remount read-only to prevent corruption.
FIX	Check dmesg for disk errors. Run fsck after unmounting. Replace failing disk if hardware error.
	<pre>dmesg grep -i 'error\ remount\ readonly' fsck -f /dev/sda1</pre>

#16 LVM Volume Group Not Found on Boot

SYMPTOM	Boot fails: Volume group not found; LVM logical volumes not activated.
ROOT CAUSE	LVM metadata corrupted, physical volume missing, or lvm2 initramfs hook missing.
FIX	Boot from rescue. Run vgscan --mknodes && vgchange -ay. Rebuild initramfs with LVM support.
	<pre>vgscan --mknodes vgchange -ay lvdisplay</pre>

#17 Partition Table Corrupted

SYMPTOM	fdisk -l shows unrecognized disk label; data inaccessible.
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ROOT CAUSE	Accidental dd, power failure during write, or hardware fault damaged partition table.
FIX	Use testdisk to scan and recover partition table. Do NOT write anything to disk first.
	<code>testdisk /dev/sda</code>

#18 Slow Disk I/O Causing System Lag

SYMPTOM	System sluggish; high iowait in top; disk reads/writes very slow.
ROOT CAUSE	Failing disk, RAID degraded, filesystem fragmentation, or I/O scheduler mismatch.
FIX	iostat -xz 1 to identify. badblocks to test disk health. Check SMART data.
	<code>iostat -xz 1 5 smartctl -a /dev/sda grep -i health</code>

#19 NFS Mount Hanging System Calls

SYMPTOM	ls or df hangs indefinitely on NFS mount; process in D (uninterruptible) state.
ROOT CAUSE	NFS server unreachable; hard mount with no timeout causes indefinite wait.
FIX	Use soft,timeo=30,retrans=3 mount options. Or unmount with umount -f -l.
	<code>umount -f -l /mnt/nfs mount -o soft,timeo=30 server:/share /mnt/nfs</code>

#20 Accidental File Deletion — Recovery

SYMPTOM	Important file deleted; need to recover before disk writes overwrite blocks.
ROOT CAUSE	rm used without backup; file removed from directory but blocks possibly still on disk.
FIX	Immediately unmount or go read-only. Use extundelete or testdisk to recover.
	<code>extundelete /dev/sda1 --restore-file /path/to/file</code>

#21 XFS Filesystem Corruption

SYMPTOM	XFS errors in dmesg; metadata I/O errors; files inaccessible.
ROOT CAUSE	Unclean shutdown, bit rot, or hardware fault corrupted XFS metadata.
FIX	xfs_repair /dev/sda1 after unmounting. Use -n for dry run first.
	<code>xfs_repair -n /dev/sda1 # dry run xfs_repair /dev/sda1 # actual repair</code>

#22 Swap Not Being Used / System OOM Despite Free Swap

SYMPTOM	System OOM kills despite swap being available; vm.swappiness=0.
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ROOT CAUSE	vm.swappiness set to 0 — kernel almost never uses swap even under memory pressure.
FIX	Set vm.swappiness=10-60 for workloads. Persist in /etc/sysctl.conf.
<pre>sysctl vm.swappiness=10 echo 'vm.swappiness=10' >> /etc/sysctl.conf</pre>	

#23 Disk UUID Changed After Clone/Replace

SYMPTOM	System fails to mount partitions after disk clone; fstab UUID not found.
ROOT CAUSE	New disk or clone has different UUID than original disk referenced in fstab.
FIX	blkid to get new UUIDs. Update /etc/fstab and /boot/grub/grub.cfg accordingly.
<pre>blkid /dev/sdb1 vim /etc/fstab</pre>	

#24 tmpfs Running Out of Space

SYMPTOM	/tmp or /run writes fail; df shows tmpfs at 100%.
ROOT CAUSE	tmpfs default size is 50% of RAM; application writing large temp files.
FIX	Increase tmpfs size: mount -o remount,size=4G /tmp. Or clean large temp files.
<pre>mount -o remount,size=4G /tmp df -h /tmp</pre>	

NETWORKING — Real-World Troubleshooting

#25 No Network After Reboot — Interface Not Coming Up

SYMPTOM	Network unreachable after reboot; ip a shows interface DOWN.
ROOT CAUSE	Network interface not configured to auto-start, or NetworkManager/netplan config missing.
FIX	Check netplan config or /etc/network/interfaces. Run netplan apply or ifup <iface>.
	<pre>netplan apply ip link set eth0 up && dhclient eth0</pre>

#26 DNS Not Resolving

SYMPTOM	ping google.com fails with unknown host; ping 8.8.8.8 works.
ROOT CAUSE	/etc/resolv.conf empty, incorrect, or overwritten by NetworkManager/DHCP.
FIX	Set nameserver in /etc/resolv.conf or configure DNS in NetworkManager/netplan.
	<pre>echo 'nameserver 8.8.8.8' > /etc/resolv.conf systemctl restart NetworkManager</pre>

#27 SSH Connection Refused

SYMPTOM	ssh user@host returns: Connection refused on port 22.
ROOT CAUSE	sshd not running, firewall blocking port 22, or sshd listening on non-standard port.
FIX	Check sshd: systemctl status sshd. Check firewall. Verify ListenAddress in sshd_config.
	<pre>systemctl start sshd ss -tlnp grep sshd ufw allow 22/tcp</pre>

#28 SSH Connection Timing Out

SYMPTOM	SSH connection hangs for 60+ seconds then refuses; no error message.
ROOT CAUSE	Reverse DNS lookup on server side blocking; or UseDNS yes in sshd_config.
FIX	Set UseDNS no in /etc/ssh/sshd_config. Or fix reverse DNS for client IP.
	<pre>echo 'UseDNS no' >> /etc/ssh/sshd_config systemctl restart sshd</pre>

#29 IP Address Conflict on Network

SYMPTOM	Intermittent connectivity; arp shows duplicate IPs; strange packet loss.
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ROOT CAUSE	Two hosts assigned same IP — static IP conflicts with DHCP range, or DHCP server bug.
FIX	arping to detect conflict. Fix static IP or exclude from DHCP pool.
	<code>arping -I eth0 192.168.1.100 ip a show eth0</code>

#30 Firewall (iptables/nftables) Blocking Traffic

SYMPTOM	Service is running and listening but requests blocked; telnet to port fails.
ROOT CAUSE	iptables DROP rule or UFW firewall blocking inbound or outbound traffic.
FIX	Check rules: iptables -L -n -v. Add allow rule or flush rules for debugging.
	<code>iptables -L -n -v ufw status verbose ufw allow 8080/tcp</code>

#31 Network Interface Flapping

SYMPTOM	Interface goes up/down repeatedly; logs show link up/link down.
ROOT CAUSE	Faulty cable/SFP, NIC driver bug, or autonegotiation failure.
FIX	Check dmesg for driver errors. Force speed/duplex: ethtool -s eth0 speed 1000 duplex full autoneg off.
	<code>dmesg grep -i 'eth0\ link\ carrier' ethtool eth0</code>

#32 Routing Table Missing Default Gateway

SYMPTOM	Cannot reach external IPs; local network works fine.
ROOT CAUSE	Default gateway not set or removed; DHCP lease expired without renewal.
FIX	Add default route: ip route add default via <gateway>. Persist via netplan/ifupdown.
	<code>ip route show ip route add default via 192.168.1.1 dev eth0</code>

#33 High Network Latency / Packet Loss

SYMPTOM	ping shows high latency or intermittent packet loss to gateway.
ROOT CAUSE	Network congestion, duplex mismatch, NIC buffer overflow, or ISP issue.
FIX	mtr for per-hop analysis. Check ethtool for errors. Tune rx/tx ring buffers.
	<code>mtr --report google.com ethtool -S eth0 grep -i error</code>

#34 Bonding/Teaming Interface Not Failing Over

SYMPTOM	One NIC fails but traffic doesn't switch to backup NIC automatically.
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ROOT CAUSE	Bonding mode misconfigured, miimon not set, or wrong active-backup mode.
FIX	Check /etc/network/interfaces or nmcli. Set miimon=100 for link detection.
	<pre>cat /proc/net/bonding/bond0 nmcli connection show bond0</pre>

#35 MTU Mismatch Causing Silent Packet Drops

SYMPTOM	Small packets work fine; large transfers (scp, rsync) hang or are extremely slow.
ROOT CAUSE	MTU size mismatch between host and network path (jumbo frames misconfigured).
FIX	Set MTU to 1500: ip link set eth0 mtu 1500. Test with ping -M do -s 1472.
	<pre>ping -M do -s 1472 -c3 8.8.8.8 ip link set eth0 mtu 1500</pre>

#36 Too Many TIME_WAIT Connections

SYMPTOM	Application cannot open new connections; ss shows thousands of TIME_WAIT sockets.
ROOT CAUSE	Short-lived TCP connections not reusing sockets; default TIME_WAIT is 60 seconds.
FIX	Enable tcp_tw_reuse and reduce fin_timeout via sysctl.
	<pre>sysctl -w net.ipv4.tcp_tw_reuse=1 sysctl -w net.ipv4.tcp_fin_timeout=15 ss -s</pre>

#37 NetworkManager Overwriting resolv.conf

SYMPTOM	Manual DNS changes to /etc/resolv.conf are reset on every DHCP renewal.
ROOT CAUSE	NetworkManager manages resolv.conf and overwrites it on network events.
FIX	Configure DNS in NetworkManager config, or mark resolv.conf immutable (chattr +i).
	<pre>nmcli con mod <conn> ipv4.dns '8.8.8.8 1.1.1.1' chattr +i /etc/resolv.conf</pre>

#38 VLAN Interface Not Coming Up

SYMPTOM	VLAN tagged interface eth0.100 missing after reboot.
ROOT CAUSE	8021q kernel module not loaded, or VLAN config not persisted in network config.
FIX	modprobe 8021q. Add VLAN config to /etc/network/interfaces or netplan.
	<pre>modprobe 8021q ip link add link eth0 name eth0.100 type vlan id 100 ip link set eth0.100 up</pre>

PROCESS & PERFORMANCE — Real-World Troubleshooting

#39 System Load High but CPU Not Busy

SYMPTOM	uptime shows load average 20+ but top shows CPU mostly idle.
ROOT CAUSE	High I/O wait — processes blocked waiting for disk or network I/O, not CPU-bound.
FIX	iostat -xz 1 to find I/O bottleneck. Identify D-state processes: <code>ps aux grep 'D'</code> .
	<code>iostat -xz 1 5 ps aux awk '\$8=="D" {print}'</code>

#40 Process in Uninterruptible Sleep (D State)

SYMPTOM	Process stuck in D state; kill -9 has no effect.
ROOT CAUSE	Process waiting for I/O that never completes — NFS hang, failing disk, or driver bug.
FIX	Identify with <code>ps aux</code> . Fix underlying I/O issue (NFS, disk). Reboot may be required.
	<code>ps aux awk '\$8=="D"' cat /proc/<pid>/wchan</code>

#41 Zombie Processes Accumulating

SYMPTOM	<code>ps aux</code> shows many processes in Z (zombie) state.
ROOT CAUSE	Parent process not calling <code>wait()</code> to reap child processes after they exit.
FIX	Identify parent: <code>ps -o ppid= <zombie_pid></code> . Fix parent app or kill parent to reap zombies.
	<code>ps aux grep 'Z' ps -o ppid= -p <zombie_pid></code>

#42 Out of Memory — OOM Killer Invoked

SYMPTOM	Random process killed; kernel log shows Out of memory: Kill process.
ROOT CAUSE	System RAM + swap exhausted; kernel OOM killer selects process to kill.
FIX	Check <code>/var/log/syslog</code> for <code>oom_score</code> . Increase RAM/swap. Set <code>oom_score_adj</code> to protect critical processes.
	<code>dmesg grep -i 'oom\\ kill' cat /proc/<pid>/oom_score</code>

#43 CPU Throttling — Thermal Throttling

SYMPTOM	System performance degrades over time; CPU frequency drops to minimum.
ROOT CAUSE	CPU overheating triggers thermal throttling to protect hardware.

FIX	Check temp: sensors. Clean heatsink/fan. Check thermal paste. Improve airflow.
	<pre>sensors watch -n1 'cat /sys/class/thermal/thermal_zone*/temp'</pre>

#44 Runaway Process Consuming All CPU

SYMPTOM	top shows single process at 100% CPU; system unresponsive.
ROOT CAUSE	Infinite loop, deadlock, or application bug causing CPU spin.
FIX	Identify PID: top or ps aux --sort=-%cpu. Kill or renice.
	<pre>ps aux --sort=-%cpu head -5 kill -15 <pid> renice -n 19 -p <pid></pre>

#45 Process Won't Die — kill -9 Not Working

SYMPTOM	kill -9 <pid> sent but process still shows in ps; not dying.
ROOT CAUSE	Process in uninterruptible D state (I/O wait); SIGKILL cannot interrupt kernel I/O wait.
FIX	Fix underlying I/O hang. Zombie processes require killing parent. D-state requires system reboot.
	<pre>cat /proc/<pid>/status grep State cat /proc/<pid>/wchan</pre>

#46 System Running Slow — Memory Pressure

SYMPTOM	System sluggish; free -h shows very low available memory; heavy swapping.
ROOT CAUSE	Memory leak in application, or too many processes competing for limited RAM.
FIX	free -h, smem, or top to find memory hogs. Kill or restart leaking process.
	<pre>free -h smem -s rss -r head -10 ps aux --sort=-%mem head -10</pre>

#47 ulimit — Too Many Open Files

SYMPTOM	Application error: Too many open files; cannot open sockets or file handles.
ROOT CAUSE	Process hitting nofile ulimit; default is often 1024 which is too low for servers.
FIX	Increase limits in /etc/security/limits.conf. Set LimitNOFILE in systemd unit.
	<pre>ulimit -n echo '* soft nofile 65536 * hard nofile 65536' >> /etc/security/limits.conf</pre>

#48 Fork Bomb — System Unresponsive Due to Process Storm

SYMPTOM	System freezes; cannot run any commands; massive number of processes.
ROOT CAUSE	Runaway script or malicious code creating unlimited child processes.

FIX	Boot to single-user mode. Kill process tree. Set nproc limit in limits.conf.
	<pre>echo '* hard nproc 1000' >> /etc/security/limits.conf killall -u <user></pre>

#49 cron Job Not Running

SYMPTOM	Scheduled cron job doesn't execute at expected time.
ROOT CAUSE	Wrong crontab syntax, script not executable, PATH not set in cron environment, or crond not running.
FIX	Test script manually. Set full paths in cron. Check /var/log/cron for errors.
	<pre>crontab -l grep CRON /var/log/syslog bash -x /path/to/script.sh</pre>

#50 systemd Service Memory Leak — Unit Restart Loop

SYMPTOM	Service using ever-increasing memory; systemd restarting it due to MemoryMax exceeded.
ROOT CAUSE	Application memory leak; or systemd MemoryMax set too low.
FIX	Set MemoryMax=512M in unit file with Restart=always. Profile app for leaks.
	<pre>systemctl show <service> grep Memory journalctl -u <service> grep -i memory</pre>

USERS & PERMISSIONS — Real-World Troubleshooting

#51 Cannot sudo — User Not in sudoers

SYMPTOM	sudo command fails: <user> is not in the sudoers file. This incident will be reported.
ROOT CAUSE	User not added to sudo group or /etc/sudoers file.
FIX	As root: <code>usermod -aG sudo username</code> . Or use visudo to add entry.
	<code>usermod -aG sudo username # OR: visudo # username ALL=(ALL:ALL) ALL</code>

#52 File Permission Denied Despite Being Owner

SYMPTOM	Permission denied accessing own file; <code>ls -la</code> shows correct ownership.
ROOT CAUSE	Execute bit not set on directory, ACL overriding permissions, or immutable flag set.
FIX	Check ACLs: <code>getfacl</code> . Check immutable: <code>lsattr</code> . Fix execute on parent directory.
	<code>getfacl /path/to/file lsattr /path/to/file chmod +x /parent/dir</code>

#53 SUID Binary Security — Unexpected Root Execution

SYMPTOM	Non-root user can execute command as root unexpectedly; security audit flags SUID bit.
ROOT CAUSE	SUID bit set on binary — any user runs it as file owner (often root).
FIX	Find SUID files: <code>find / -perm -4000</code> . Remove SUID where not needed: <code>chmod u-s /path</code> .
	<code>find / -perm -4000 -type f 2>/dev/null chmod u-s /usr/bin/suspicious</code>

#54 User Cannot Log In — Account Locked

SYMPTOM	SSH or console login fails: Account temporarily locked or Permission denied.
ROOT CAUSE	Too many failed login attempts (<code>pam_tally2/faillock</code>), or account manually locked.
FIX	Unlock: <code>faillock --reset --user <user></code> . Or: <code>passwd -u <user></code> .
	<code>faillock --user <username> faillock --reset --user <username></code>

#55 PAM Authentication Failure

SYMPTOM	Login fails with Authentication failure even with correct password.
ROOT CAUSE	PAM module misconfigured in /etc/pam.d/, SSSD/LDAP not reachable, or shadow file corruption.

FIX	Check /etc/pam.d/common-auth. Test with pamtester. Check SSSD status.
	<pre>pamtester login <user> authenticate systemctl status sssd</pre>

#56 umask Causing Wrong File Permissions

SYMPTOM	Newly created files have wrong permissions; group cannot read/write shared files.
ROOT CAUSE	User's umask set too restrictive (e.g., 077 instead of 022).
FIX	Set correct umask in ~/.bashrc or /etc/profile: umask 022.
	<pre>umask umask 022 echo 'umask 022' >> /etc/profile</pre>

#57 sudo Password Prompt Timing Out

SYMPTOM	sudo fails immediately with sorry, a password is required — no prompt shown in scripts.
ROOT CAUSE	sudo requiretty enabled in sudoers; or TTY not available in non-interactive context.
FIX	Add Defaults !requiretty in /etc/sudoers for service accounts. Use NOPASSWD for automation.
	<pre>visudo # Add: Defaults:username !requiretty</pre>

#58 ACL Entries Ignored After Copy

SYMPTOM	ACLs correctly set on files but disappear after cp or rsync.
ROOT CAUSE	cp and rsync don't preserve ACLs by default.
FIX	Use cp -a or rsync -aAX (capital A for ACLs, X for xattrs).
	<pre>rsync -aAX /source/ /destination/ getfacl /destination/file</pre>

#59 Shared Directory Write Access Issues

SYMPTOM	Members of group cannot write to shared directory despite being in correct group.
ROOT CAUSE	Directory missing group write bit, sticky bit issues, or setgid not set.
FIX	chmod g+w dir. Set setgid: chmod g+s dir so new files inherit group.
	<pre>chmod g+ws /shared/dir chown :groupname /shared/dir</pre>

#60 Home Directory Not Created for New User

SYMPTOM	New user can log in but no home directory exists; shell errors on login.
ROOT CAUSE	useradd used without -m flag; or /etc/skel missing.

FIX	Create home dir manually: mkhomedir_helper username. Or use adduser which creates it automatically.
<pre>mkhomedir_helper username # Or use: adduser username</pre>	

#61 User Cannot Change Password — PAM Policy Rejection

SYMPTOM	passwd returns: Password fails quality checking policy or BAD PASSWORD.
ROOT CAUSE	PAM pam_pwquality/cracklib enforcing password complexity requirements.
FIX	Check /etc/pam.d/common-password and /etc/security/pwquality.conf for policy settings.
<pre>cat /etc/security/pwquality.conf pamtester passwd <user> chauthtok</pre>	

#62 SSH Key Authentication Not Working

SYMPTOM	SSH key login fails; falls back to password or Permission denied (publickey).
ROOT CAUSE	Wrong permissions on ~/.ssh or authorized_keys; SELinux context wrong; key format issue.
FIX	chmod 700 ~/.ssh; chmod 600 ~/.ssh/authorized_keys. Check sshd log for specific error.
<pre>chmod 700 ~/.ssh && chmod 600 ~/.ssh/authorized_keys tail -f /var/log/auth.log</pre>	

PACKAGE MANAGEMENT — Real-World Troubleshooting

#63 apt-get update Fails — GPG Key Error

SYMPTOM	apt-get update returns: NO_PUBKEY or The following signatures couldn't be verified.
ROOT CAUSE	Repository GPG key missing or expired.
FIX	Add missing key: apt-key adv --keyserver keyserver.ubuntu.com --recv-keys <KEYID>.
	<pre>apt-key adv --keyserver keyserver.ubuntu.com --recv-keys <KEY_ID> apt-get update</pre>

#64 Broken Package Dependencies

SYMPTOM	apt-get install fails: unresolved dependencies or held broken packages.
ROOT CAUSE	Partial upgrade, manually installed conflicting package, or PPA conflict.
FIX	apt --fix-broken install. Or dpkg --configure -a. Check dpkg -l for half-installed.
	<pre>apt --fix-broken install dpkg --configure -a apt-get install -f</pre>

#65 yum/dnf Transaction Error — RPM Database Corrupt

SYMPTOM	yum or dnf fails: rpmdb: damaged header or RPM database corruption.
ROOT CAUSE	Power failure during transaction or concurrent rpm access corrupted RPM database.
FIX	Rebuild RPM database: rpm --rebuilddb. Or remove and recreate /var/lib/rpm/__db.*.
	<pre>rm -f /var/lib/rpm/__db.* rpm --rebuilddb dnf clean all</pre>

#66 Package Held Back During apt upgrade

SYMPTOM	apt upgrade shows N packages not upgraded; critical updates not applied.
ROOT CAUSE	Packages held with apt-mark hold, or phased updates, or dependency conflict.
FIX	Check held packages: apt-mark showhold. Use apt full-upgrade for dist-upgrade logic.
	<pre>apt-mark showhold apt full-upgrade</pre>

#67 pip Install Fails — SSL Certificate Error

SYMPTOM	pip install fails: SSL: CERTIFICATE_VERIFY_FAILED.
ROOT CAUSE	Corporate proxy intercepting HTTPS with untrusted cert, or system CA bundle outdated.

FIX	Pass --trusted-host or install cert into system trust store.
	<pre>pip install --trusted-host pypi.org --trusted-host files.pythonhosted.org <package></pre>

#68 Snap Package Confined — Permission Denied

SYMPTOM	Snap-installed app cannot access files outside home directory or /media.
ROOT CAUSE	Snap strict confinement; app needs interface connections for extra permissions.
FIX	Connect required interfaces: snap connect <snap>:home. Or use --devmode for testing.
	<pre>snap connections <snap-name> snap connect <snap>:removable-media</pre>

#69 Kernel Not Updating — linux-image Held Back

SYMPTOM	New kernel available but linux-image package not upgrading.
ROOT CAUSE	Disk space issue in /boot, or package held, or grub not configured to update.
FIX	Clean old kernels: apt autoremove --purge. Check /boot space: df -h /boot.
	<pre>df -h /boot apt autoremove --purge apt-get install linux-image-generic</pre>

#70 Package Manager Lock File — Another Process Using apt

SYMPTOM	apt-get fails: Could not get lock /var/lib/dpkg/lock-frontent.
ROOT CAUSE	Another apt/dpkg process running (or crashed and left lock).
FIX	Wait for existing process. If stale: remove lock files and restart.
	<pre>lsof /var/lib/dpkg/lock-frontent rm /var/lib/dpkg/lock-frontent apt-get install -f</pre>

LOGS & MONITORING — Real-World Troubleshooting

#71 journald Logs Consuming Too Much Disk

SYMPTOM	/var/log/journal using gigabytes; disk filling up.
ROOT CAUSE	SystemMaxUse not configured in journald.conf; logs accumulate indefinitely.
FIX	Set SystemMaxUse=500M in /etc/systemd/journald.conf. Vacuum old logs.
	<pre>journalctl --disk-usage journalctl --vacuum-size=500M echo 'SystemMaxUse=500M' >> /etc/systemd/journald.conf</pre>

#72 rsyslog Not Writing to Log File

SYMPTOM	Expected log messages not appearing in /var/log/syslog or custom log file.
ROOT CAUSE	rsyslog service not running, wrong facility/severity filter, or file permission issue.
FIX	Check rsyslog status. Verify rules in /etc/rsyslog.conf. Check target file permissions.
	<pre>systemctl status rsyslog rsyslogd -N1 # config test tail -f /var/log/syslog</pre>

#73 Log Rotation Not Working — Logs Growing Unbounded

SYMPTOM	Log files in /var/log growing to gigabytes; logrotate not running.
ROOT CAUSE	logrotate cron job not running, config syntax error, or wrong log file path in config.
FIX	Test: logrotate -d /etc/logrotate.conf. Force: logrotate -f. Fix config errors.
	<pre>logrotate -d /etc/logrotate.conf # dry run logrotate -vf /etc/logrotate.d/nginx</pre>

#74 dmesg Buffer Overflowing — Losing Kernel Messages

SYMPTOM	Important kernel messages disappearing; dmesg truncated.
ROOT CAUSE	Kernel ring buffer too small for high-volume messages (busy network/storage).
FIX	Increase ring buffer: dmesg -s 1048576. Or configure klogd/rsyslog to persist kernel logs.
	<pre>dmesg -s 1048576 tail -50 echo 'kern.* /var/log/kernel.log' >> /etc/rsyslog.conf</pre>

#75 auditd Not Logging Security Events

SYMPTOM	Expected audit events (file access, sudo, login) not appearing in /var/log/audit/audit.log.
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ROOT CAUSE	auditd not running, audit rules not loaded, or audit log partition full.
FIX	Start auditd. Load rules: auditctl. Check audit log disk usage.
	<pre>systemctl start auditd auditctl -l auditctl -w /etc/passwd -p wa -k passwd_changes</pre>

#76 System Time Drift Causing Log Correlation Issues

SYMPTOM	Logs from different servers have timestamps that don't align; root cause analysis confused.
ROOT CAUSE	NTP not synchronizing; different servers using different NTP sources.
FIX	Ensure all servers use same NTP source. Use chronyc tracking to verify sync.
	<pre>chronyc tracking chronyc sources -v timedatectl set-ntp true</pre>

#77 Syslog Flooding With Repeated Messages

SYMPTOM	/var/log/syslog growing rapidly; same message repeated thousands of times per minute.
ROOT CAUSE	Application or kernel module in error loop generating excessive log output.
FIX	Identify source with grep. Enable RepeatedMsgReduction in rsyslog. Fix root cause application.
	<pre>grep -c 'repeated message' /var/log/syslog tail -f /var/log/syslog grep -v <noisy_process></pre>

#78 SNMP Not Responding on Host

SYMPTOM	Monitoring system cannot poll SNMP; snmpwalk from external host times out.
ROOT CAUSE	snmpd not running, community string wrong, firewall blocking UDP 161, or ACL in snmpd.conf.
FIX	Check snmpd status. Verify community string. Allow UDP 161 in firewall.
	<pre>systemctl status snmpd snmpwalk -v2c -c public localhost ufw allow 161/udp</pre>

SECURITY & HARDENING — Real-World Troubleshooting

#79 SELinux Blocking Application

SYMPTOM	Application fails with Permission denied despite correct Linux permissions; audit.log shows AVC denied.
ROOT CAUSE	SELinux policy doesn't allow the operation; context mismatch on file or port.
FIX	Check: ausearch -m avc. Use audit2allow to generate policy. Or restorecon to fix context.
<pre>ausearch -m avc -ts recent audit2why < /var/log/audit/audit.log restorecon -Rv /path</pre>	

#80 AppArmor Profile Denying Access

SYMPTOM	Application denied access to file/network; /var/log/syslog shows apparmor='DENIED'.
ROOT CAUSE	AppArmor profile in enforce mode restricting access not listed in profile.
FIX	Check: aa-status. Put profile in complain mode to log without blocking while fixing.
<pre>aa-status aa-complain /etc/apparmor.d/usr.bin.myapp aa-logprof</pre>	

#81 SSH Brute Force Attempts — fail2ban Not Blocking

SYMPTOM	auth.log flooded with SSH brute force attempts; fail2ban not banning IPs.
ROOT CAUSE	fail2ban not running, wrong log path in jail.conf, or filter regex not matching.
FIX	Check fail2ban status. Verify jail.local config. Test filter: fail2ban-regex.
<pre>fail2ban-client status sshd fail2ban-regex /var/log/auth.log /etc/fail2ban/filter.d/sshd.conf</pre>	

#82 Unauthorized Cron Job Found

SYMPTOM	Security audit finds cron jobs not placed by admins; potential persistence mechanism.
ROOT CAUSE	Compromised account or malware adding cron entries for persistence.
FIX	Audit all crontabs: crontab -l -u <user>. Check /etc/cron.* and /var/spool/cron.
<pre>for u in \$(cut -f1 -d: /etc/passwd); do crontab -u \$u -l 2>/dev/null; done ls /etc/cron.d/</pre>	

#83 World-Writable Files Found — Security Risk

SYMPTOM	Security scan finds world-writable files or directories outside of /tmp.
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ROOT CAUSE	Incorrect chmod applied; scripts run with too-permissive umask.
FIX	Find and fix: <code>find / -xdev -perm -o+w -not -path '/proc/*' -not -path '/tmp/*'</code> .
	<pre>find / -xdev -perm -o+w -not -path '/proc/*' -not -path '/tmp/*' 2>/dev/null chmod o-w /path/to/file</pre>

#84 SSL Certificate Expired on Service

SYMPTOM	HTTPS service showing certificate expired error; clients rejecting connection.
ROOT CAUSE	TLS certificate not renewed before expiry; Let's Encrypt certbot renewal failed.
FIX	Renew: certbot renew. Check systemd timer: <code>systemctl status certbot.timer</code> . Fix renewal config.
	<pre>certbot renew --dry-run certbot certificates openssl x509 -in /etc/ssl/cert.pem -noout -dates</pre>

#85 Suspicious Network Connections Found

SYMPTOM	Security audit shows unexpected outbound connections to unknown IPs.
ROOT CAUSE	Compromised service, malware, or misconfigured application making unauthorized connections.
FIX	<code>ss -tulpn</code> to see listeners. <code>netstat -antp</code> for connections. Kill suspicious processes.
	<pre>ss -tulpn netstat -antp grep ESTABLISHED lsof -i -P -n grep -v LISTEN</pre>

#86 sysctl Hardening Settings Not Persisting

SYMPTOM	Security sysctl settings applied with <code>sysctl -w</code> reset after reboot.
ROOT CAUSE	Settings applied at runtime but not persisted in <code>/etc/sysctl.conf</code> or <code>/etc/sysctl.d/</code> .
FIX	Write settings to <code>/etc/sysctl.d/99-security.conf</code> and run <code>sysctl --system</code> .
	<pre>echo 'net.ipv4.ip_forward=0' > /etc/sysctl.d/99-security.conf sysctl -- system</pre>

#87 Password Hashes Visible in /etc/passwd

SYMPTOM	Security audit finds password hash in <code>/etc/passwd</code> instead of x in password field.
ROOT CAUSE	Shadow passwords not enabled; system using old-style <code>/etc/passwd</code> password storage.
FIX	Run <code>pwconv</code> to migrate hashes to <code>/etc/shadow</code> .
	<pre>pwconv chmod 640 /etc/shadow ls -la /etc/shadow</pre>

#88 Core Dump Containing Sensitive Data

SYMPTOM	Core dump files in /tmp or /var/crash containing memory with credentials.
ROOT CAUSE	Core dumps enabled (ulimit -c unlimited) and process crashes dump memory to disk.
FIX	Disable core dumps: ulimit -c 0. Or configure core_pattern to safer location.
<pre>ulimit -c 0 echo '* hard core 0' >> /etc/security/limits.conf echo 'kernel.core_pattern=/bin/false' >> /etc/sysctl.conf</pre>	

ADVANCED & KERNEL — Real-World Troubleshooting

#89 Kernel Module Not Loading

SYMPTOM	Device or feature not working; lsmod doesn't show expected module.
ROOT CAUSE	Module blacklisted in /etc/modprobe.d/, or missing from initramfs, or kernel version mismatch.
FIX	Load manually: modprobe <module>. Check blacklist. Rebuild initramfs if needed.
<pre>modprobe <module_name> lsmod grep <module> cat /etc/modprobe.d/blacklist.conf</pre>	

#90 Kernel Module Blacklist Preventing Hardware Function

SYMPTOM	NIC, GPU, or audio device not working; relevant module shows in blacklist.
ROOT CAUSE	Driver conflict or stability issue caused module to be blacklisted.
FIX	Remove from /etc/modprobe.d/blacklist.conf. Reboot or modprobe to load.
<pre>grep <module> /etc/modprobe.d/*.conf modprobe <module></pre>	

#91 IRQ Conflicts Causing Device Issues

SYMPTOM	Hardware device unstable; dmesg shows IRQ conflicts or spurious IRQs.
ROOT CAUSE	Two devices sharing IRQ in non-APIC mode; legacy ISA IRQ assignment issue.
FIX	Enable APIC in BIOS. Check: cat /proc/interrupts. Reassign IRQs via BIOS or irqbalance.
<pre>cat /proc/interrupts service irqbalance start</pre>	

#92 Huge Pages Not Configured for Database

SYMPTOM	Database (Oracle/PostgreSQL) performing poorly; memory not using huge pages.
ROOT CAUSE	Transparent huge pages disabled or huge pages not pre-allocated for the workload.
FIX	Enable THP or allocate huge pages: sysctl vm.nr_hugepages=512.
<pre>cat /sys/kernel/mm/transparent_hugepage/enabled sysctl -w vm.nr_hugepages=512 grep Huge /proc/meminfo</pre>	

#93 Kernel Soft Lockup Detected

SYMPTOM	dmesg shows: BUG: soft lockup - CPU#X stuck for Xs! or hard lockup.
ROOT CAUSE	CPU stuck in kernel code for too long — driver bug, deadlock, or overloaded CPU.

FIX	Update kernel and drivers. Check for hardware issues. Capture full stack trace.
	<code>dmesg grep -i lockup journalctl -k grep -i lockup</code>

#94 Memory Fragmentation Causing Allocation Failures

SYMPTOM	dmesg shows: page allocation failure; order:4; application crashes despite free memory.
ROOT CAUSE	Physical memory fragmented; large contiguous pages unavailable even though total free is high.
FIX	Defragment memory: echo 1 > /proc/sys/vm/compact_memory. Or reduce order of allocation in app.
	<pre>echo 1 > /proc/sys/vm/compact_memory cat /proc/buddyinfo</pre>

#95 eBPF Program Failing to Load

SYMPTOM	BPF tool (bpftool, Cilium, etc.) fails: Operation not permitted or verifier error.
ROOT CAUSE	Kernel BPF JIT disabled, insufficient capabilities, or eBPF program fails verifier.
FIX	Enable BPF JIT: sysctl net.core.bpf_jit_enable=1. Run as root or with CAP_BPF.
	<pre>sysctl -w net.core.bpf_jit_enable=1 bpftool prog list bpftool tracepoint:*</pre>

#96 NUMA Imbalance Causing Performance Issues

SYMPTOM	Multi-socket server has high latency for some processes; numastat shows imbalanced memory.
ROOT CAUSE	Processes not NUMA-aware; accessing memory on remote NUMA node causing latency.
FIX	Bind process to NUMA node: numactl --cpunodebind=0 --membind=0 <cmd>.
	<code>numastat numactl --hardware numactl --cpunodebind=0 --membind=0 myapp</code>

#97 CPU Frequency Scaling Not Working

SYMPTOM	CPU stays at minimum frequency; cpufreq governor not applying performance mode.
ROOT CAUSE	Wrong cpufreq governor set; cpupower tools not installed; BIOS power management overriding.
FIX	Set governor: cpupower frequency-set -g performance. Check BIOS power settings.
	<code>cpupower frequency-info</code> <code>cpupower frequency-set -g performance</code> <code>cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_governor</code>

#98 Crash Dump (kdump) Not Capturing Kernel Panic

SYMPTOM	After kernel panic, no vmcore file in /var/crash; crash not captured.
ROOT CAUSE	kdump not enabled, crash kernel memory not reserved, or kexec not loaded.
FIX	Enable kdump: systemctl enable kdump. Reserve memory: add crashkernel=auto to GRUB.
<pre>systemctl enable --now kdump grep crashkernel /proc/cmdline kdumpctl status</pre>	

#99 GRUB Password Not Set — Security Risk

SYMPTOM	Security audit finds GRUB boot menu accessible without password; anyone can edit boot params.
ROOT CAUSE	GRUB password protection not configured.
FIX	Set GRUB password: grub-mkpasswd-pbkdf2. Add to /etc/grub.d/40_custom. Run update-grub.
<pre>grub-mkpasswd-pbkdf2 # Add to /etc/grub.d/40_custom: # set superusers='root' # password pbkdf2 root <hash></pre>	

#100 Strace — Debugging Unknown Process Failures

SYMPTOM	Application fails intermittently with no useful error; logs show nothing helpful.
ROOT CAUSE	Underlying syscall failing silently; missing file, network error, or resource limit hit.
FIX	Attach strace to running process or run under strace to see all syscalls.
<pre>strace -p <pid> -e trace=file,network 2>&1 head -50 strace -o /tmp/trace.log -ff myapp</pre>	